

MATHEMATICS
Introduction to Modern Algebra I
Quiz 4.

1) If G is a group such that $a^2 = e$ for all $a \in G$, show that G is abelian.

Proof. Let $a, b \in G$, then $abab = (ab)^2 = e$ and $a^2 = b^2 = e$. But then $a^{-1} = a^{-1}a^2 = a$, $b^{-1} = b^{-1}b^2 = b$ and so $ab = b^{-1}a^{-1}abab = b^{-1}a^{-1}e = b^{-1}a^{-1} = ba$.

2) If G is a group, show that the identity element is unique and show that inverses are unique.

Proof. Let $e, f \in G$ be identity elements so that $eg = g = ge$ and $fg = g = gf$ for any $g \in G$. Then $e = fe = f$.

Let $a \in G$ and suppose that $b, c \in G$ are inverses of a so that $ab = ba = e$ and $ac = ca = e$. Then $c = ec = bac = be = b$.